Course AI Assistant

CST3492 — Java Enterprise Application Development

Group C — Team 12

Team Members

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Plugin Features

Course Assistance

- PDF/PPTX course materials
- Answers with source evidence

Code Context

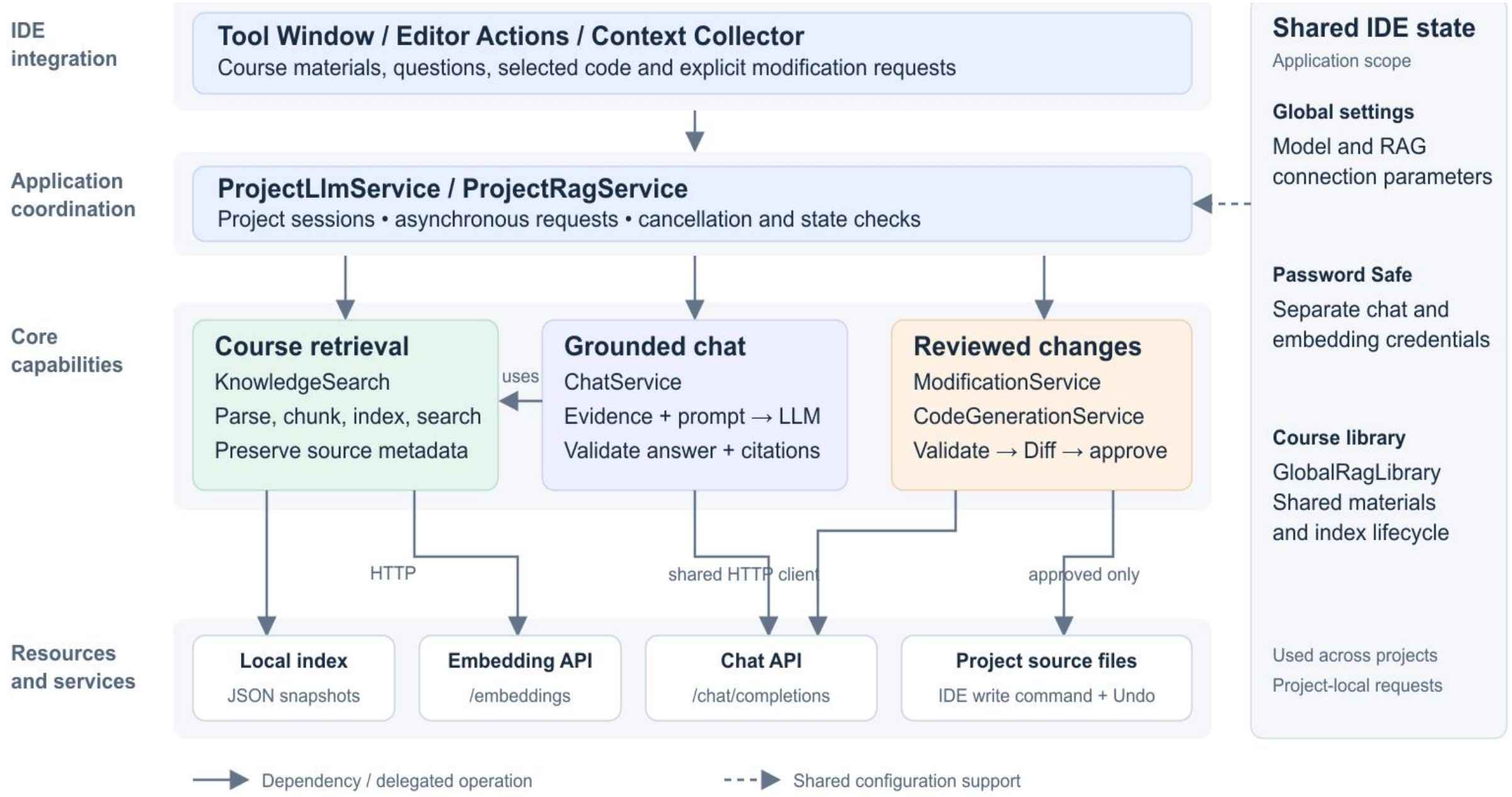
- Select Java code
- Ask about the selection

Reviewed Changes

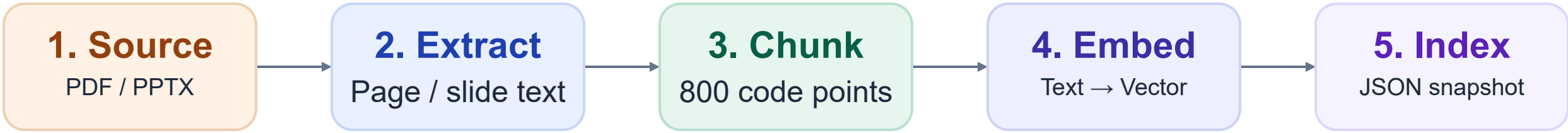
- Generate a proposal
- Review, approve or cancel

F0: Course-grounded assistance
F2: Reviewed single-file changes

System Architecture



Course Material Indexing



Chunking Strategy

- Preserve paragraph/sentence boundaries
- Maximum 800 code points per chunk
- Overlap budget: up to 120

Source Traceability

- Retain document and page/slide
- PDF text: Apache PDFBox
- PPTX text: Apache POI

Knowledge Storage and Retrieval

File Registry

- Original file paths
- Corpus ID / format version

Chunk + Vector + Source

- Text + vector
- Document + page/slide

Index Manifest

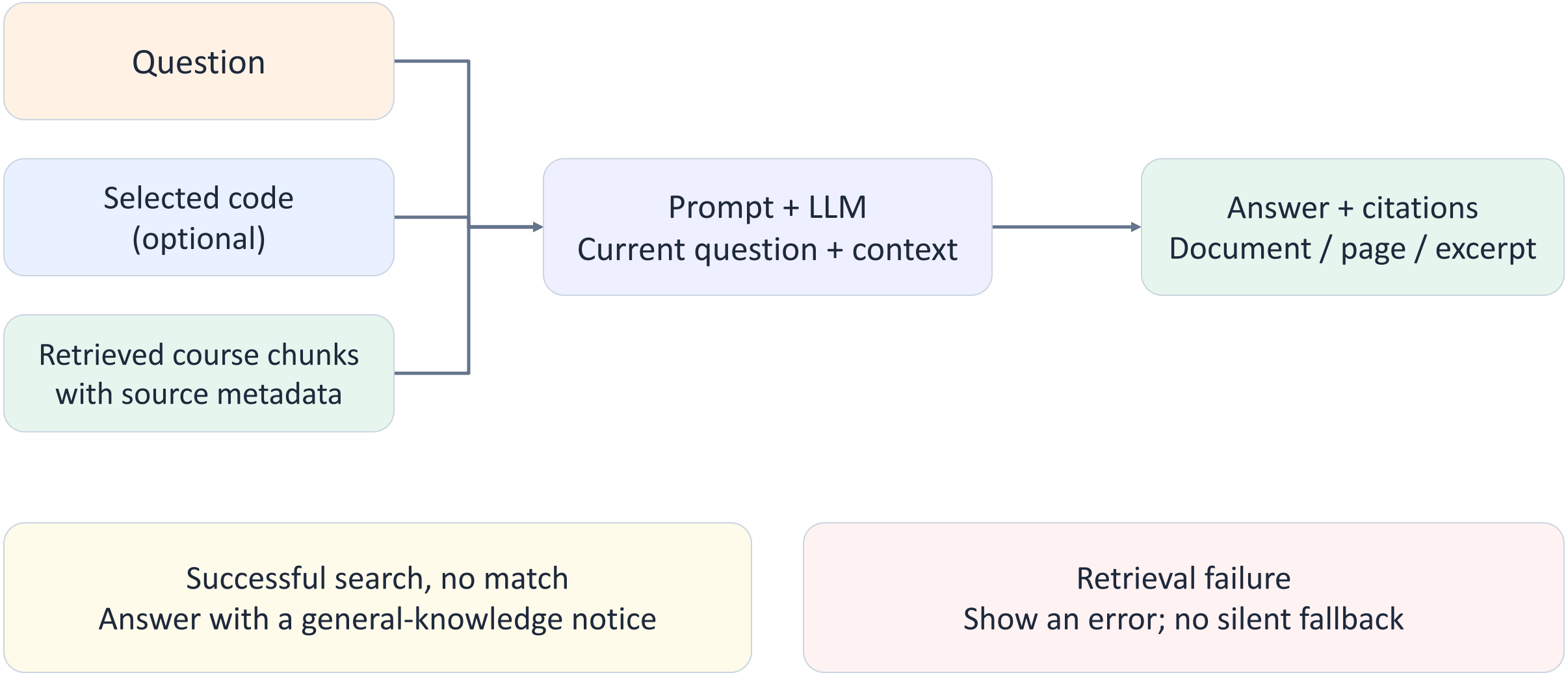
- Model/config fingerprints
- Snapshot integrity

Retrieval Flow

Question → Embedding → Similarity search → Filter → Up to 5 chunks

- Reuse stored document vectors
- Local vector store saved as JSON
- Default threshold: 0.70; not calibrated

Course Q&A Workflow



LLM Integration and Source Validation

Prompt Rules

- Use supplied evidence
- Treat code/evidence as data
- Return structured answer + IDs

Local Checks

- Validate current evidence IDs
- Match body markers to citation list
- Invalid citations → response error

Citation Chain: Source → Display

Source Metadata
doc + page + excerpt

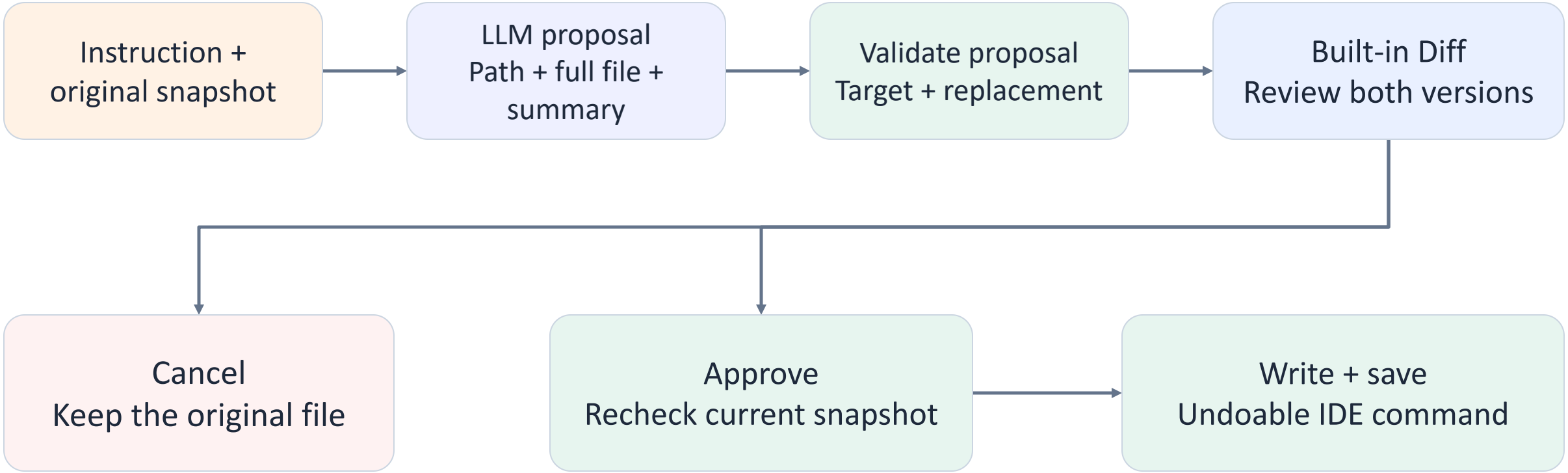
Evidence ID
Supplied reference ID

Validated Citation
Validate ID + markers

Visible Marker
[1]

Valid reference IDs do not prove that every claim is supported.

Reviewed Code Modification



One Java file per proposal. Generation and preview never write source files.

Design Decisions and Efficiency

Problem	Design Response	Effect
Repeated parsing/embedding	Persist and reuse vectors	Reduce repeated work
Slow requests / stale replies	Background work + state checks	Keep IDE responsive
File edited during review	Recheck snapshot before write	Avoid overwriting edits

Additional Efficiency Measures

- Shared settings and course library across projects
- Request sessions remain project-local

Team Collaboration

Module Responsibilities

杜冰焱	UI, editor context and integration
黄茂军	Document pipeline and retrieval
李一芃	LLM chat and code generation
董泽福	Diff review and controlled writes

Integration Approach

Shared contracts

↓

Feature branches

↓

Incremental integration

Limitations and Future Work

Current Limitation

No OCR for image-only materials

↓

Future Direction

Add text recognition

Current Limitation

Exhaustive local vector search

↓

Future Direction

Scalable local indexing

Current Limitation

Generated code not automatically compiled

↓

Future Direction

Optional compile checks

Traceable course answers.

Reviewed single-file changes.

Questions?